

Violation Webhook API Documentation (Draft)

1. Base Configuration

Base URL

https://welappsuat.welspun.com

Authentication

All webhook APIs support authentication using the following header:

x-webhook-auth: <API_AUTH_KEY> (U2blpNYCc8cIdS2ZpNd7)

Common Headers

Header	Required	Description
Content-Type	Yes	Request content type
Accept	Yes	Expected response format
x-webhook-auth	Yes	Webhook authentication key

Note: Requests without a valid x-webhook-auth header will be rejected.

Content-Type

application/json

2. PPE Violation Webhook

Endpoint

POST /webhooks/report_ppe_violation

Description

Receives PPE violation detections from AI systems.

A single request may contain multiple PPE detections for the same person and frame.

One database record is created per detection object.

Request Headers

Header	Required
Content-Type	Yes
Accept	Yes
x-webhook-auth	Yes

Request Body

```
{
  "frame": {
    "frame_id": 56234,
    "cam_id": 27,
    "timestamp": "2026-04-22T18:21:09",
    "frame_url": "https://storage.example.com/ppe/frame_56234.jpg"
  },
  "person": {
    "person_id": 11845,
    "first_seen": "2026-04-20T17:05:33",
    "total_violations": 2,
    "crop_s3_url": "https://storage.example.com/ppe/person_11845.jpg"
  },
  "detections": [
    {
      "id": 201,
      "frame_id": 56234,
      "person_id": 11845,
      "class_name": "NO-Safety-Vest",
      "confidence": 0.97,
      "x1": 320,
      "y1": 180,
      "x2": 500,
      "y2": 420
    }
  ]
}
```

Field Definitions

Frame

Field	Type	Mandatory	Description
frame_id	number	Yes	Frame identifier
cam_id	number	No	Camera identifier
timestamp	datetime	No	Detection timestamp
frame_url	string	No	Frame image URL

Person

Field	Type	Mandatory	Description
person_id	number	Yes	Person identifier
first_seen	datetime	No	First detected timestamp
total_violations	number	No	Total violations count
crop_s3_url	string	No	Person crop image URL

Detection

Field	Type	Mandatory	Description
id	number	Yes	Detection identifier
class_name	string	Yes	PPE violation type
confidence	decimal	No	Detection confidence
x1	number	No	Bounding box coordinate
y1	number	No	Bounding box coordinate
x2	number	No	Bounding box coordinate
y2	number	No	Bounding box coordinate

Validation Rules

Mandatory Fields

- frame.frame_id
- person.person_id
- detections

Detection Validation

- detections must contain at least one object.

Duplicate Validation

System validates uniqueness of:

- frame_id
- detection_id

Duplicate records are rejected.

Success Response

```
{
  "success": true,
  "message": "2 PPE violation(s) logged successfully",
  "frame_id": 56234,
  "person_id": 11845
}
```

Error Response

```
{
  "success": false,
  "message": "frame_id, person_id and detections are required"
}
```

Sample CURL Request

```
curl --location '{{violation_module_base_url}}/webhooks/report_ppe_violation' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'x-webhook-auth: {{API_AUTH_KEY}}' \
--data @payload.json
```

3. Site Insight / Defect Webhook

Endpoint

POST /webhooks/report_site_insight

Description

Receives site inspection and defect analysis data from external inspection systems.

A single request may contain multiple defects.

One database record is created per defect.

Request Headers

Header	Required
Content-Type	Yes
Accept	Yes
x-webhook-auth	Yes

Request Body

```
{
  "inspection_id": 40876,
  "project_id": 9,
  "site": {
    "site_id": 15,
    "name": "North Quarry Zone",
    "gps": "21.1458,79.0882",
    "status": "defect_found",
    "description": "Cracks observed on retaining wall near excavation area."
  },
  "structure": {
    "structure_id": 31,
    "type": "Retaining Wall",
    "gps": "21.1461,79.0885",
    "coords_on_map": "528,142",
    "status": "major",
    "description": "Visible surface cracks and concrete spalling detected."
  },
  "images": [],
  "defects": [],
  "map_s3_url": "https://s3.example.com/site_map.jpg"
}
```

Field Definitions

Root Fields

Field	Type	Mandatory	Description
inspection_id	number	Yes	Inspection identifier
project_id	number	No	Project identifier
map_s3_url	string	No	Site map image URL

Site

Field	Type	Mandatory	Description
site_id	number	No	Site identifier
name	string	No	Site name
gps	string	No	GPS coordinates
status	string	No	Site status
description	string	No	Site remarks

Structure

Field	Type	Mandatory	Description
structure_id	number	No	Structure identifier
type	string	No	Structure type
gps	string	No	GPS coordinates

Field	Type	Mandatory	Description
coords_on_map	string	No	Position on site map
status	string	No	Structure status
description	string	No	Structure remarks

Images

Field	Type	Mandatory	Description
image_id	number	No	Image identifier
image_url	string	No	Image URL
captured_at	datetime	No	Capture timestamp

Defects

Field	Type	Mandatory	Description
defect_id	number	Yes	Defect identifier
image_id	number	No	Related image identifier
defect_npz_url	string	No	NPZ model file URL
3d_model_coords	object	No	Defect coordinates in 3D model

Validation Rules

Mandatory Fields

- inspection_id

Duplicate Validation

System validates uniqueness of:

- inspection_id
- defect_id

Duplicate defect records are rejected.

Defect Processing

- Each defect creates a separate database record.
- Matching image information is automatically linked using image_id.

Success Response

```
{
  "success": true,
  "message": "2 defect record(s) logged successfully",
  "inspection_id": 40876
}
```

Error Response

```
{
  "success": false,
  "message": "inspection_id is required"
}
```

Sample CURL Request

```
curl --location '{{violation_module_base_url}}/webhooks/report_site_insight' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'x-webhook-auth: {{API_AUTH_KEY}}' \
--data @payload.json
```

General Error Response

```
{
  "success": false,
  "message": "Failed to process request"
}
```

4. Vehicle Detection Webhook

Endpoint

POST /webhooks/report_vehicle_detection

Description

Receives vehicle detection events from external AI/CCTV systems and stores vehicle movement, plate recognition, vehicle classification, load analysis, and associated metadata.

Each request creates a single vehicle detection record.

Request Headers

Header	Required
Content-Type	Yes
Accept	Yes
x-webhook-auth	Yes

Request Body

```
{
  "detection_id": 10582,
  "project_id": 7,
  "cam_id": 21,
  "camera_location": "Plant Area C - Exit Gate",
  "timestamp": "2026-06-10T14:45:30",
  "direction": "exit",
  "vehicle": {
    "morth_class": 6,
    "plate_number": "GJ05XY9876",
    "confidence": {
      "vehicle": 0.97,
      "plate": 0.93,
      "ocr": 0.95,
      "morth_class": 0.91
    }
  },
  "bounding_boxes": {
    "vehicle": {
      "x": 385,
      "y": 102,
      "w": 245,
```

```
    "h": 170
  },
  "plate": {
    "x": 465,
    "y": 228,
    "w": 95,
    "h": 30
  }
},
"image_urls": {
  "full_frame": "https://s3.example.com/frame.jpg",
  "vehicle_crop": "https://s3.example.com/vehicle.jpg",
  "load_frame": "https://s3.example.com/load.jpg"
},
"load_analysis": {
  "load_status": "normal",
  "load_type": "sand",
  "total_volume_m3": 10.0,
  "load_volume_m3": 8.7
}
}
```

Field Definitions

Field	Type	Mandatory	Description
detection_id	number	Yes	Unique detection identifier
project_id	number	Yes	Project identifier
cam_id	number	No	Camera identifier
camera_location	string	No	Camera location
timestamp	datetime	No	Detection timestamp
direction	string	No	Vehicle direction (entry, exit)
vehicle.morth_class	number	No	MORTH vehicle classification
vehicle.plate_number	string	No	Vehicle registration number
vehicle.confidence	object	No	Detection confidence scores
bounding_boxes	object	No	Vehicle and plate coordinates
image_urls	object	No	Image URLs generated by AI engine
load_analysis	object	No	Load analysis information
load_analysis.load_status	string	No	Example: normal, overloaded, underloaded
load_analysis.load_type	string	No	Material type
load_analysis.total_volume_m3	decimal	No	Total truck volume
load_analysis.load_volume_m3	decimal	No	Actual detected load volume

Validation Rules

Mandatory Fields

- detection_id
- project_id

Duplicate Validation

System validates uniqueness of:

- detection_id

Duplicate records are rejected.

Numeric Validation

Following fields must be numeric when provided:

- project_id
- cam_id
- morth_class
- total_volume_m3
- load_volume_m3

Success Response

```
{
  "success": true,
  "message": "Vehicle detection logged successfully",
  "id": 15,
  "detection_id": 10582
}
```

Error Response

```
{
  "success": false,
  "message": "detection_id and project_id are required"
}
```

Sample CURL Request

```
curl --location '{{violation_module_base_url}}/webhooks/report_vehicle_detection' \
--header 'Content-Type: application/json' \
--header 'Accept: application/json' \
--header 'x-webhook-auth: {{API_AUTH_KEY}}' \
--data '{
  "detection_id": 10582,
  "project_id": 7,
  "cam_id": 21,
  "camera_location": "Plant Area C - Exit Gate",
  "timestamp": "2026-06-10T14:45:30",
  "direction": "exit"
}'
```

Notes

- Vehicle Detection creates one record per request.
- PPE Violation creates one record per detection.
- Site Insight creates one record per defect.
- Duplicate records based on unique database constraints are rejected.
- All timestamps should be sent in ISO-8601 format.
- All image URLs should be publicly accessible or accessible to the receiving system.